Microcomputer Control for Sensorless Brushless Motor

KENICHI IIZUKA, HIDEO UZUHASHI, MINORU KANO, TSUNEHIRO ENDO, AND KATSUO MOHRI

Abstract—**The microcomputer control of a brushless motor without a shaft position sensor is studied. A method which permits the determination of the permanent magnet rotor position by the back electromotive force (EMF) induced in the stator windings and the starting technique which uses the motor as a synchronous motor at standstill are explained. The motor voltage is chopped by commutator transistors to change the motor speed. The control system consists of a 4-b single-chip microcomputer and two quad-comparators.**

INTRODUCTION

TO MODULATE the capacity of room air conditioners continuously in proportion to the load results in energy saving and a comfortable environment. It is necessary to develop a variable-speed motor built in a compressor for such a type of air conditioner. The speed of the brushless motor with a permanent magnet rotor can be easily controlled over a wide range by changing the motor voltage. Moreover, the motor has high efficiency with magnet excitation. Nevertheless, this type of motor needs a rotor position sensor, and this reduces the system ruggedness and complicates the motor configuration. In particular, for the motor built in a completely sealed compressor, mechanical shaft position sensing is judged to be difficult to apply and to be impossible to mass produce due to 1) the sensor low reliability in high-temperature and refrigerant environments and 2) the need of a extra hermetic terminal of the sensor signal lead wires.

The purpose of this paper is to study a control method which permits the determination of the rotor position by the back EMF to eliminate this sensor. Also, since the back EMF is zero at standstill, a starting technique which permits the starting of a separately controlled synchronous motor without a sensor is described.

The control strategy of this system is shown in Fig. 1. The motor speed control and elimination of a mechanical sensor (sensorless) are the main points of this system. For the terminal voltage sensing circuit which substitutes a mechanical sensor, three low-pass filters and three voltage comparators are provided. The performance of the starting at standstill and motor speed control using a chopper are achieved by the programmed software in the microcomputer.

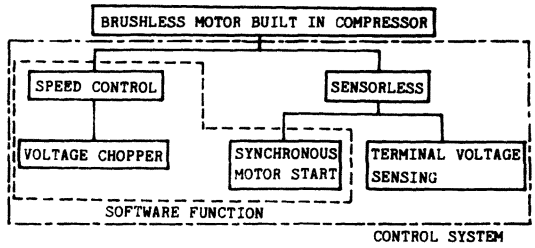


Fig. 1. Control strategy.

For this control system, a 4-b single-chip microcomputer is adopted because 1) of the high-speed operation—all instructions are excuted in 2 μs; 2) it contains the programable logic array (PLA) which protects the inverter from damage when the program goes down; 3) low price for mass production, 2000 × 8-b ROM, 128 × 4-b RAM, 1 μs instruction time, and a 42-pin dual in-line package are the computer basic elements.

MICROCOMPUTER CONTROL SYSTEM

A. Control Function

The system block diagram is shown in Fig. 2. The brushless motor comprises a three-phase star-connected stator and a four-pole permanent magnet rotor. A 120-electrical-degree type inverter is used as a commutator of the motor. The inverter whose control capacity is 1.5 kVA comprises six n-p-n type Darlington transistors and six flywheel diodes. The control circuit is divided into four parts: the microcomputer, the position sensing circuit, the current detector and the inverter drive circuit. The total number of LSI's is three; the computer and two quad voltage comparators. The software functions of the computer are as follows.

1) Starting at Standstill: The computer generates six inverter drive pulses and starts the motor as a separately controlled synchronous one. Accelerating the frequency of the pulses at a sufficiently high speed, enough back EMF from the motor is available to drive the brushless motor. The computer changes the stage from a synchronous motor to a brushless one commutated by the terminal voltage sensor output signals.

2) Steady-State Drive: The computer receives three voltage sensor output signals and logically converts them into six inverter drive signals.

3) Speed Control: The computer generates a pulsewidth modulation (PWM) signal and varies the on-duty cycle of the pulse in accordance with the speed command. The motor voltage is chopped by commutator transistors to change the motor speed. The inverter transistors are used both as the

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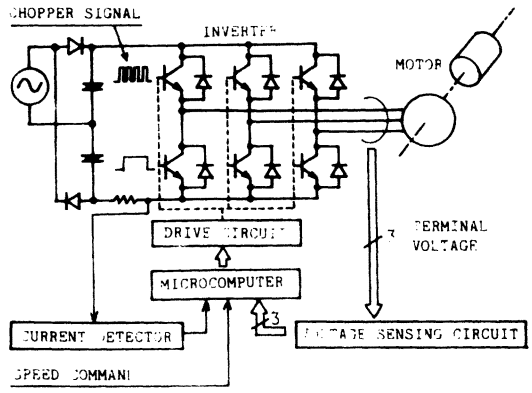


Fig. 2. Motor drive system.

commutator of the motor and as the voltage chopper for the speed control.

4) Protection of Overload: To protect power devices from damage, the computer decreases the on-duty cycle of the PWM pulse signal when overcurrent flows.

5) Display the Causes of Abnormal Motor Stop: When something is wrong with the motor operation, the computer stops the inverter drive and displays the cause of motor stop with four LED's.

B. Terminal Voltage Sensing

In a three-phase brushless motor and a 120-electrical-degree type inverter, transistor commutation occurres at intervals of 60 electrical degrees. This causes important transient in the motor terminal voltages, so some techniques are necessary for the direct use of these voltages for sensing rotor position.

The method of terminal voltage sensing is as follows: Three terminal voltages are VA, VB, and VC. The neutral voltage is VN, as shown in Fig. 3 (a). The three back EMF's and three impedances of the stator windings are Ea, Eb, Ec, and Za, Zb, Zc, respectively. The inverter source dc voltage is Vo. The terminal voltages are three-phase trapezoidal waveforms each with a phase angle, of 120°. Take the case of the phase A terminal voltage VA, for example, as shown in Fig. 3 (b). The motor current passes through A winding during the period from 60 to 180 and from 240 to 360 electrical degrees. No current passes through A winding during the period from 0 to 60 and from 180 to 240 electrical degrees. An equivalent circuit is shown in Fig. 3 (c) during the period from 180 to 240 electrical degrees.

As the transistor forward voltage drop is very small, each terminal voltage is

$$\begin{aligned} VA &= VN + Ea \\ VB &= Vo \\ VC &= 0, \end{aligned} \tag{1}$$

and the motor current I and the neutral voltage VN are

$$\begin{aligned} I &= \frac{Vo - (Eb - Ec)}{Zb + Zc} \\ VN &= Vo - Eb - I \times Zb. \end{aligned} \tag{2}$$

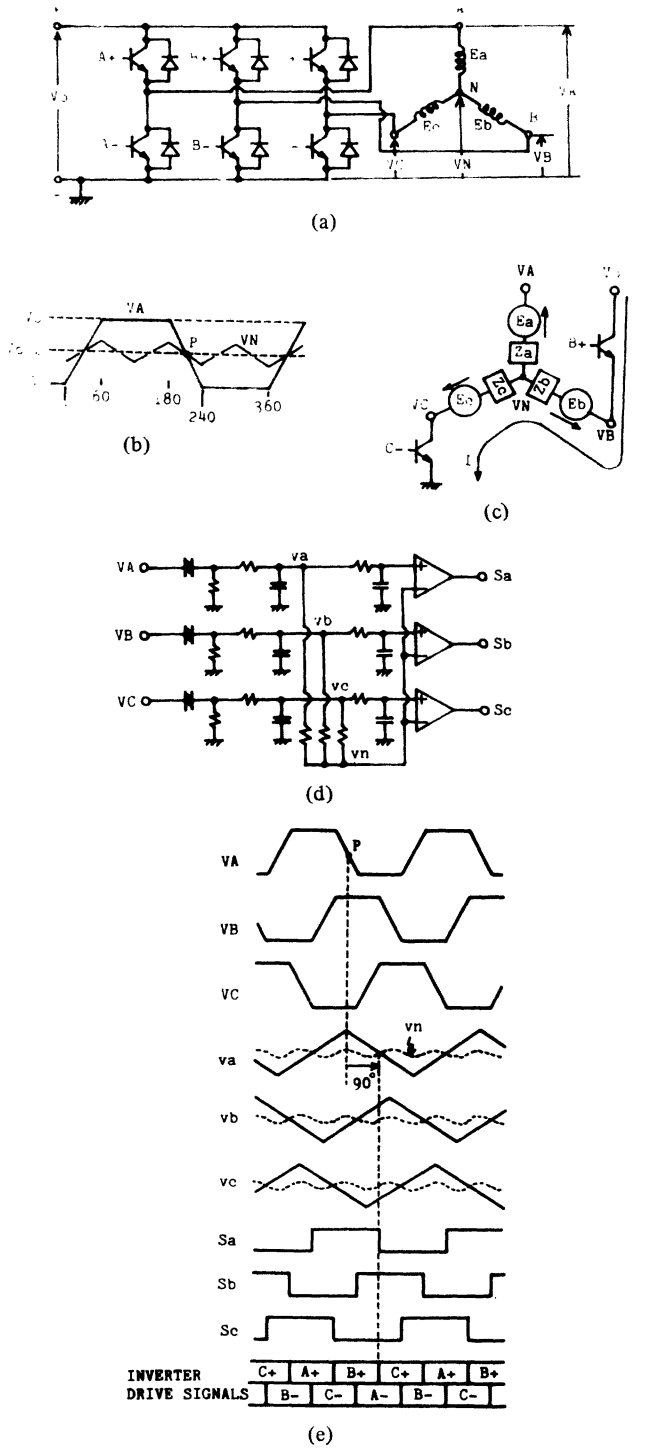


Fig. 3. Terminal voltage sensing. (a) Inverter configuration. (b) Terminal voltage waveform. (c) Equivalent circuit. (d) Low-pass filter. (e) Time chart.

In general, the back EMF is symmetrical and the impedance of windings are equal to one another, hence

$$\begin{aligned} Ea &= f(\theta) \\ Eb &= f(\theta - 2\pi/3) \\ Ec &= f(\theta - 4\pi/3) \\ Z &= Za = Zb = Zc. \end{aligned} \tag{3}$$

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From (1)–(3), the phase A terminal voltage is

$$VA = VN + Ea = \frac{Vo - Eb - Ec}{2} + Ea. \quad (4)$$

If $Ea = 0$, then from (2), $Eb = -Ec$, hence

$$VA = VN = Vo/2. \quad (5)$$

Equation (5) indicates that $VA = VN$ at the point P shown in Fig. 3 (b) when $Ea = 0$. This position of the point P is independent of the load current. The terminal voltages of phase B and C also have the same relation between VA and VN.

It is necessary to delay the point P 90 electrical degrees to use this point directly as the inverter drive signals. So, a circuit including three low-pass filters is provided for the 90 electrical degrees phase delay of each terminal voltage as shown in Fig. 3(d).

Fig 3(e) shows the time chart of this circuit. The terminal voltages VA, VB, and VC are converted into the triangular waveform signals va, vb, and vc through the low-pass filters, and va, vb, and vc are delayed 90 electrical degrees as compared with VA, VB, and VC, respectively. The resistor star-connected circuit whose terminal voltages are va, vb, and vc derives vn at the neutral point which is delayed the same electrical degrees as compared with VN. Comparing the voltage signals va with vn, vb with vn, and vc with vn, form the pulse signals Sa, Sb, and Sc, respectively, by the voltage comparators. Finally, the computer converts the signals Sa, Sb, and Sc into six inverter drive signals $A+$ $C-$ logically. The low-pass filter cutoff frequency is designed over the entire speed range.

Fig. 4 shows the terminal voltage VA and filtered waveform of VA, va. VA is chopped by the commutator transistors to change the motor speed. Therefore, VA contains the chopper frequency. Nevertheless, these commutation transients contained in terminal voltages are eliminated by the low-pass filters. Moreover, the p-p voltage level of the filtered waveform signals va, vb, and vc varies little with the speed because the average of the terminal voltage is proportional to the motor speed for brushless motors.

C. Starting at Standstill

Since the back EMF is zero at standstill, separately controlled signals of a synchronous motor are generated in the computer. The frequency of these signals is accelerated by degrees, and when the motor back EMF is sufficient to allow the determination of the rotor position by the terminal voltages, the computer changes the operation stage from a separately controlled synchronous motor to a self-controlled brushless one, as shown in Fig. 5.

Nevertheless, this change of the stage does not go well, and the motor steps out, if there is the large phase angle difference between the voltage sensing signals (Sa, Sb, and Sc) and the synchronous drive signals generated in the computer. In order to prevent the motor from stepping out, the computer monitors the phase angle difference between both signals. When the difference is equal to or smaller than the designed constant

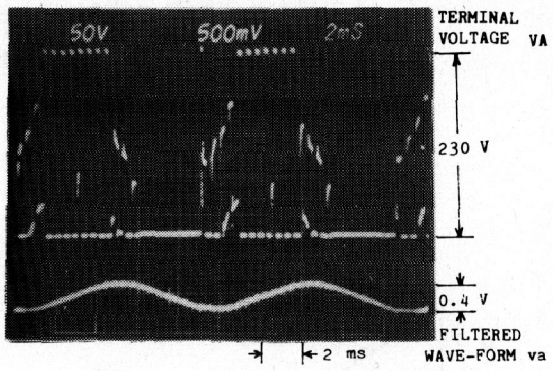


Fig. 4. Waveforms of VA and va.

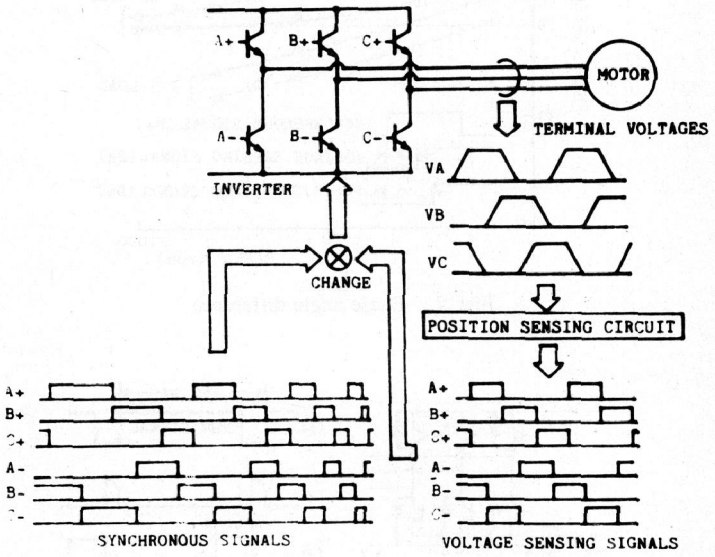


Fig. 5. Starting as a synchronous motor.

difference, the computer changes the motor stage, as shown in Fig. 6.

The computer generates the synchronous signals as follows. The motor speed is N, the number of the rotor pole is P, the period of the synchronous signal is Ti, then

$$N = \frac{120}{P \times Ti} \quad \text{r/min.}$$

Ti is decreased as a function of time. Ti is converted into Xi, by using the standard time ts

$$Tn = ts \times Xn$$

$$Xn = (Xn - 1) - (Xn - 1)/16.$$

The computer calculates the Xn and accelerates the motor.

As the phase angle difference δ between the voltage sensing signal and the synchronous signal is dependent on the motor voltage, the load, and the starting current, the mechanical shaft position sensor is installed on trial on the motor to determine the appropriate δ, as shown in Fig. 7.

The phase angle difference δ between the synchronous signal $B+$ and the voltage sensing signal Sa (shown in Fig. 3(e)) and the phase angle difference γ between the signal $B+$ and the mechanical shaft sensor output signal Da are shown in Fig. 7 as a function of the synchronous motor speed. Changing

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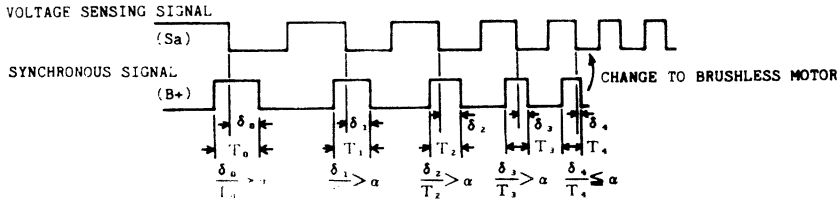


Fig. 6. Change to brushless motor.

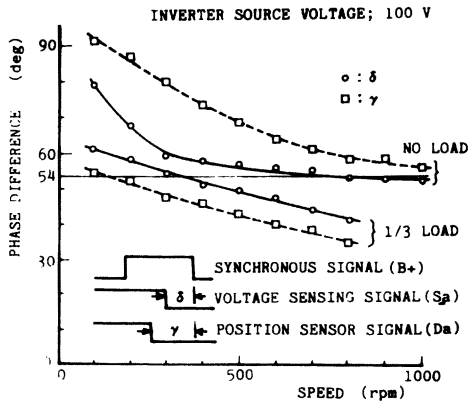


Fig. 7. Phase angle difference.

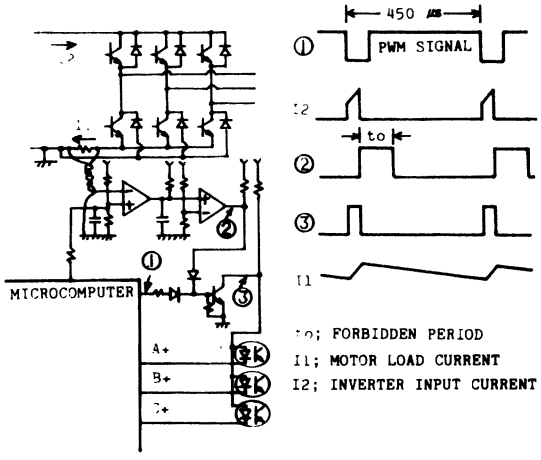


Fig. 8. Current limitter.

the motor stage is forbidden both below 200 r/min because of the transient rotor vibration and above 1000 r/min because of the stepping out by variation of the load. Therefore, the appropriate δ is determined about 54° with 5-A current limitter.

The starting current limitation method at standstill is shown in Fig. 8. The motor load current $I1$ but not the inverter input current $I2$ is detected by a resistor. The upper three transistors are driven by the PWM chopper signal. The resistor is provided between the lower three transistors and three flywheel diodes. The current limitter which comprises two voltage comparators sets the inverter operation forbidden period (to) after detecting the setup overcurrent. It is difficult to limit the current precisely because of the circuit delay time and the motor voltage variation, if the inverter input current $I2$ is used.

D. Speed Control

Fig. 9 shows the speed control block diagram of the software. Six synchronous signals generator provides the inverter drive signals through the PLA when the motor starts at standstill. Monitoring the phase difference angle δ between the synchronous signal ($B+$) and the voltage sensing signal (Sa), the generator accelerates the frequency of the synchronous signals. The computer changes the motor stage to the brushless motor when δ is below the appropriate phase angle. The starting current limitter puts into no operation after changing the stage.

The computer receives the serial speed command and the three voltage sensing signals, then it accelerates the motor speed until the command speed. The motor speed control method is to chop the motor voltage by the commutator transistors which are driven by the PWM signal generator. The generator controls on-duty cycle of the PWM signal in accordance with the speed difference between the command speed and the measured speed.

The flow of this computer control is shown in Fig. 10 with the problem analysis diagram (PAD). The program consists of the main routine, the input interrupt routine, and the timer interrupt routine. Display of the cause of abnormal motor stop, preparing the motor start, starting as a synchronous motor, speed acceleration, and steady-state motor drive are achieved in the main routine. The input interrupt routine is triggered with the instantaneous overcurrent, then the computer cuts off the inverter drive signals in 14 μs. The timer interrupt routine generates the PWM signal whose frequency is 2.2 kHz. The performance of the main routine and this timer interrupt routine is achieved alternately every 450 μs (2.2 kHz).

Software configuration of the PWM signal generator is shown in Fig. 11. As it takes 60 μs to prepare the counter-1 operation, the method of time idling of CPU is used when the period of off-duty or on-duty cycle for the PWM signal is below 60 μs. The counter-2 whose standard clock pulse is provided from the PWM signal generator measures the motor speed.

Fig. 12 shows the computer operational time. The computer must execute each function within the time β which is the period of the transient time among the voltage sensing signals, so each function is achieved in serial order.

EXPERIMENTAL RESULTS

The phase difference angle δ is shown in Fig. 13 as a function of the time after starting. The motor changes the stage at 600 r/min in case of no load and at 300 r/min in case of one-third of the rating load. It takes about 2.8 s to change the stage after starting.

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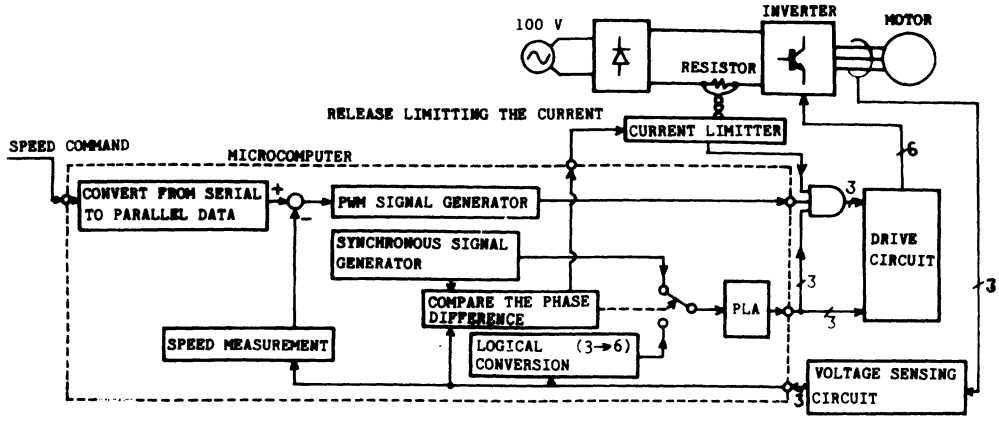


Fig. 9. Block diagram for speed control.

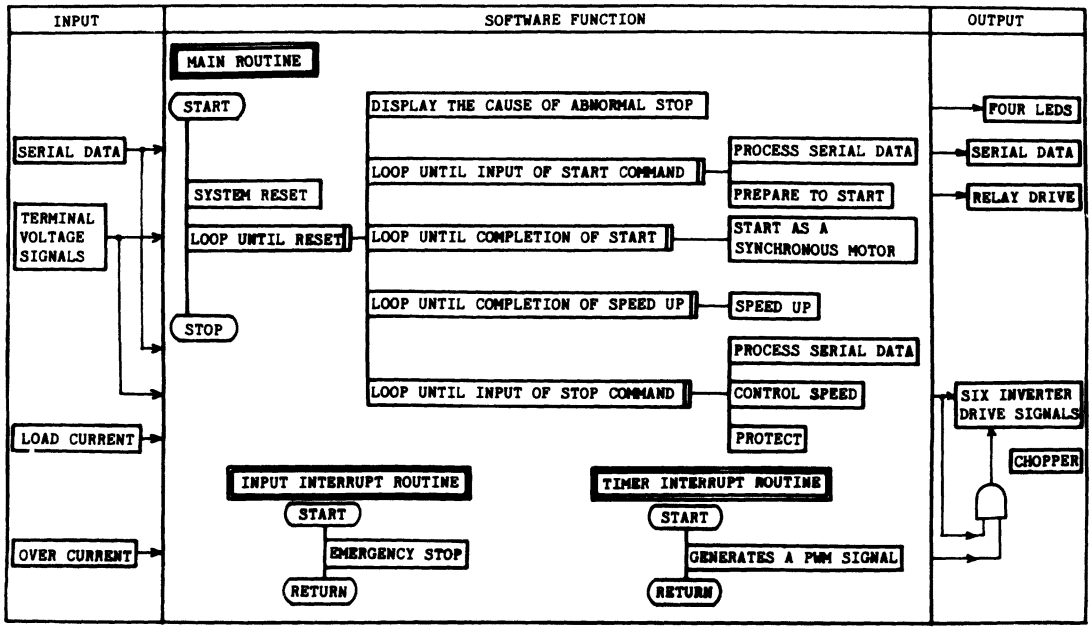


Fig. 10. Control flow.

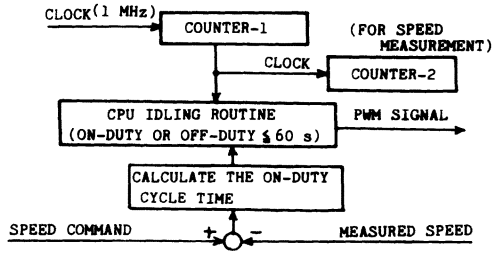


Fig. 11. PWM signal generator.

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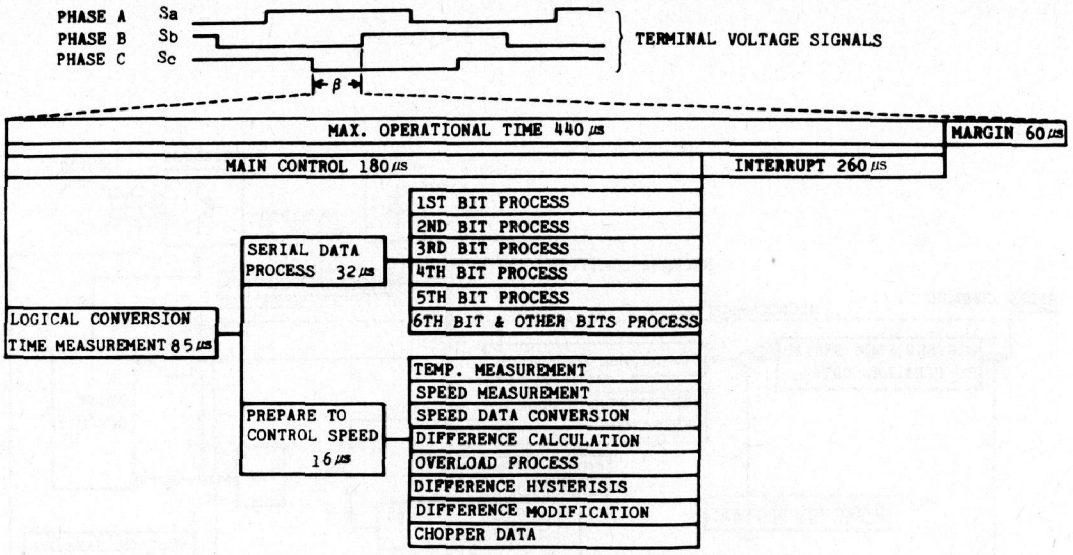


Fig. 12. Operational time.

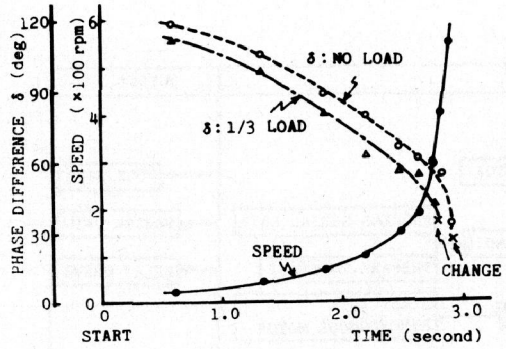
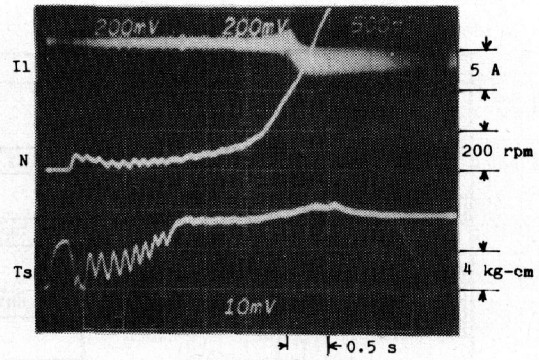


Fig. 13. Acceleration and change stage.



I1; LIMITTED CURRENT (A)
N; SPEED (rpm)
Ts; STARTING TORQUE (kg-cm)

(a)

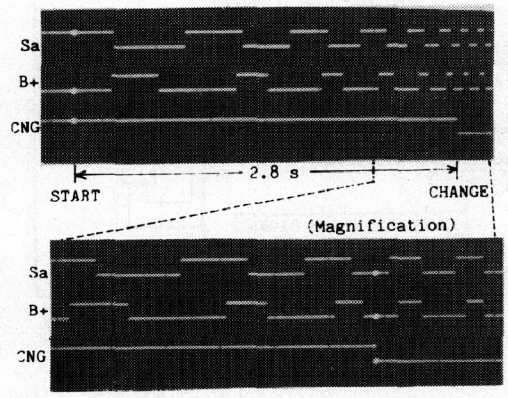
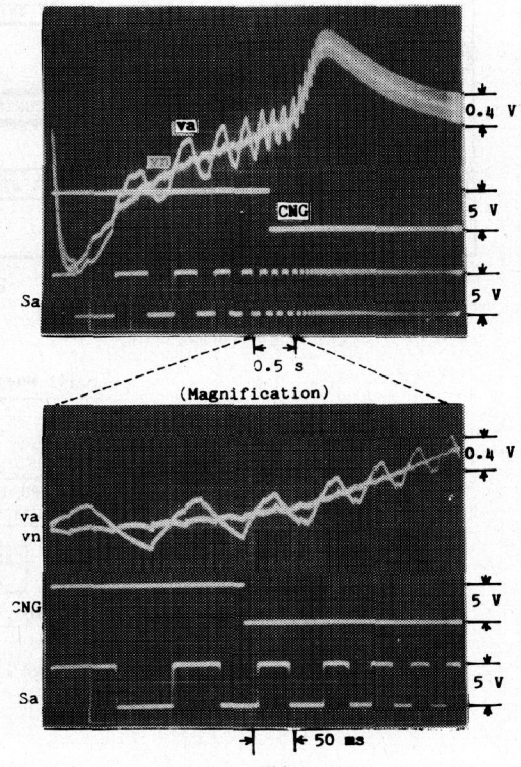


Fig. 14. Transient pulse at starting.



(b)

Fig. 15. Transient motor element at starting. (a) Transient motor element. (b) Filtered terminal voltage waveform.

Fig. 14 shows the transient pulse signals at starting. *Sa* is the voltage sensing signal. *B+* is the synchronous signal, and CNG is the stage changing signal.

Fig. 15(a) shows the transient motor elements at starting. The motor current *I*1 is limited to about 5 A. *N* is the motor speed. *Ts* is the starting torque, one-third of the rating load (7 kg·cm).

The filtered terminal voltage waveform, *va* and *vn* at starting are shown in Fig. 15(b). As accelerated ratio of the motor speed is large in time of changing the stage, the dc component of *va* varies rapidly. Nevertheless, the output *Sa* compared *va* with *vn* through the voltage comparator is not

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influenced by the variation, because the dc component of vn varies as the same level as that of va.

CONCLUSION

The microcomputer control of a brushless motor without a shaft position sensor has been studied. The method of the motor terminal voltage sensing at steady-state operation has been proposed. The starting at standstill has been realized by the technique which uses the brushless motor as a separately controlled synchronous one. This technique needs no particular device. The PWM signal generated in the computer has chopped the motor voltage by the commutator transistors to control the motor speed.

The whole control system needs a 4-b single-chip microcomputer and two quad-comparators. The proposed control system has been used for the drive of brushless motor in compressors.

Room air conditioners which contain these control system have been mass-produced since 1982. There has been no particular problem for these system since then.

APPENDIX

The rating specifications of the test brushless motor are

power	1.2 kW
speed	5700 r/min
torque	20 kg·cm
voltage	180 V
current	7.2 A

The speed range is from 1950 to 5700 r/min and excitation from ferrite permanent magnet (four poles).

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